

# Module/Pathway/Taxon Review:

## Mitochondrial Complex I Core (N & Q Modules) in *Pseudomonas putida* KT2440

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**Review topic:** Mitochondrial respiratory Complex I core (NADH:ubiquinone oxidoreductase) — catalytic N and Q modules (NDUFV1/NDUFV2/NDUFS1-3/NDUFS6-8/NDUFA9) **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00190` "Oxidative phosphorylation" (44 primary genes; 54 candidates) **Module area:** energy\_respiration\_inorganic\_metabolism

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### 1. Executive Summary

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The catalytic core of mitochondrial respiratory Complex I — the seven conserved core subunits distributed across the NADH-oxidising **N-module** and the ubiquinone-reducing **Q-module** — is **fully present and satisfiable** in *Pseudomonas putida* KT2440, but it is encoded by the **bacterial** `nuo` operon (**NDH-1**) rather than by any eukaryote-style `NDUF*` genes. Every one of the seven conserved catalytic core subunits has a clean 1:1 bacterial ortholog in the KT2440 candidate list: the N-module subunits NDUFV1, NDUFV2 and NDUFS1 map to **nuoF (PP\_4123)**, **nuoE (PP\_4122)** and **nuoG (PP\_4124)**; the Q-module subunits NDUFS7 (PSST) and NDUFS8 (TYKY) map to **nuoB (PP\_4120)** and **nuoI (PP\_4126)**; and the two cofactor-less Q-module core subunits NDUFS2 (49 kDa) and NDUFS3 (30 kDa) are both encoded by the **single fused** `nuoCD` **gene (PP\_4121)**, exactly as in *Escherichia coli* and other gamma-proteobacteria.

The three **accessory/supernumerary** subunits in the review scope — **NDUFS4**, **NDUFS6** and **NDUFA9** — are **eukaryote-specific additions** to Complex I and have no bacterial orthologs. Their absence from the KT2440 candidate list is expected and correct; they should be marked `not_expected_in_target_taxon`, not flagged as gaps. Bacterial NDH-1 is the "minimal" ~14-subunit enzyme, roughly half the size of the 44-subunit mammalian complex, and lacks the ~30 supernumerary subunits that arose in the eukaryotic lineage.

For curation, the central conclusions are: (i) the module is **covered** at the catalytic core level but **needs revision** because its generic boundary is eukaryote-centric and expects `NDUF*` gene names that do not exist in bacteria; (ii) the KEGG `ppu00190` bucket is a **whole-OXPHOS conflation** and must not be treated as a Complex I gene set — only PP\_4119–PP\_4131 (nuoA–N) belong to Complex I, while the other ~40 candidates are Complexes II–V and terminal oxidases; and (iii) the `ndh` gene (**PP\_0626**) is a **type-II NADH dehydrogenase (NDH-2)**, a non-homologous, non-proton-pumping single-subunit alternative that must not be mapped onto any Complex I core step. KT2440 carries a **single-copy, complete, syntenic** `nuo` locus with no paralogous second cluster, so satisfiability is unambiguous. All conclusions rest on strong homology/orthology evidence with robust biochemical transfer from *E. coli* and *Thermus thermophilus*; no KT2440-specific subunit-mutant data were located.

## 2. Target-Organism Pathway Definition

### 2.1 What process is in scope

The reviewed module is the **catalytic peripheral (hydrophilic) arm core of Complex I / NDH-1** — specifically the electron-input **N-module** (FMN + entry Fe-S clusters, NADH oxidation) and the electron-output **Q-module** (terminal Fe-S wire + quinone reduction). In *P. putida* KT2440 the physiological enzyme is the bacterial **proton-translocating NADH:quinone oxidoreductase (NDH-1)**, encoded by the `nuo` operon. It oxidises NADH, passes electrons through FMN and a chain of iron-sulfur clusters to reduce the native quinone pool, and couples this to proton translocation across the inner membrane.

The core fold, cofactor set and electron-transfer logic of the peripheral arm are **highly conserved from bacteria to mammals** (PMID: 41977177), which is exactly what makes the bacterial enzyme the accepted "minimal model" for the mitochondrial complex (PMID: 26807915).

### 2.2 Neighboring pathways to keep separate

The KEGG map `ppu00190` "Oxidative phosphorylation" is an **overview bucket for all five OXPHOS complexes plus accessory energetics**, not a Complex I module. For this review it must be decomposed:

| Complex / system                                       | KT2440 genes in the bucket                                                                                                                                            | In scope for this module?           |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Complex I (NDH-1)</b>                               | <code>nuoA-N</code> = PP_4119–PP_4131                                                                                                                                 | <b>YES (core only)</b>              |
| Type-II NADH dehydrogenase (NDH-2)                     | <code>ndh</code> = PP_0626                                                                                                                                            | No — non-homologous alternative     |
| Complex II (SDH)                                       | <code>sdhABCD</code> = PP_4190–4193                                                                                                                                   | No (TCA/<br><code>ppu00020</code> ) |
| Complex III (cytochrome <i>bc<sub>1</sub></i> )        | <code>petABC</code> = PP_1317–1319                                                                                                                                    | No                                  |
| Complex IV / terminal oxidases                         | <code>cox</code> / <code>ctaD</code> (PP_0103–0106), <code>cyoABCD</code> (PP_0812–0815), <code>ccoNOQP-I/II</code> (PP_4250–4258), <code>cioAB</code> (PP_4650–4651) | No                                  |
| Complex V (F <sub>1</sub> F <sub>0</sub> ATP synthase) | <code>atpA-H</code> = PP_5412–5419                                                                                                                                    | No                                  |
| Broad flavoprotein                                     | PP_2867 (pyridine nucleotide-disulphide oxidoreductase)                                                                                                               | No — not a Nuo subunit              |

Neighboring maps to keep distinct: TCA cycle (`ppu00020`), polyphosphate metabolism (`ppu03018`), porphyrin/heme biosynthesis (`ppu00860`), and photosynthesis-family electron transport (`ppu04148`).

## 2.3 Alternate names and database definitions

- **NDH-1 / Complex I / NADH:ubiquinone oxidoreductase / NADH-quinone oxidoreductase** all refer to the same `nuo`-encoded enzyme.
- Bacterial subunit names use the **Nuo** nomenclature (NuoA–N); *T. thermophilus* uses **Nqo1–14**; mammals use **NDUF\*** and **ND1–6**. The mapping across nomenclatures is standard (see §3).
- **EC** for the complex: 7.1.1.2 (formerly 1.6.5.3); individual KT2440 subunits are annotated EC 7.1.1.-.



## 4. Candidate Genes and Evidence

### 4.1 Complex I core subunits (in scope, high confidence)

| Gene                  | Locus   | UniProt | Mito ortholog / module role            | Module | Evidence type                               | Call                 |
|-----------------------|---------|---------|----------------------------------------|--------|---------------------------------------------|----------------------|
| <a href="#">nuoF</a>  | PP_4123 | Q88FH3  | NDUFV1 — FMN/<br>NADH catalytic        | N      | Homology +<br>biochem transfer              | Covered              |
| <a href="#">nuoE</a>  | PP_4122 | Q88FH4  | NDUFV2 — N1a<br>[2Fe-2S]               | N      | Homology                                    | Covered              |
| <a href="#">nuoG</a>  | PP_4124 | Q88FH2  | NDUFS1 — 75<br>kDa Fe-S                | N      | Homology                                    | Covered              |
| <a href="#">nuoB</a>  | PP_4120 | Q88FH6  | NDUFS7 (PSST)<br>— N2 [4Fe-4S]         | Q      | Homology +<br>structural                    | Covered              |
| <a href="#">nuoCD</a> | PP_4121 | Q88FH5  | NDUFS3 +<br>NDUFS2 (fused C/<br>D)     | Q      | Homology +<br>explicit fusion<br>annotation | Covered<br>(2 steps) |
| <a href="#">nuoI</a>  | PP_4126 | Q88FH0  | NDUFS8 (TYKY)<br>— N6a/N6b<br>[4Fe-4S] | Q      | Homology                                    | Covered              |

Supporting membrane-arm and connector subunits also present (not part of the N/Q catalytic-core scope but confirming a complete enzyme): [nuoA](#) (PP\_4119), [nuoH](#) (PP\_4125, ND1), [nuoJ](#) (PP\_4127), [nuoK](#) (PP\_4128), [nuoL](#) (PP\_4129, ND5), [nuoM](#) (PP\_4130, ND4), [nuoN](#) (PP\_4131, ND2). All lie in a contiguous, canonically ordered operon PP\_4119→PP\_4131.

**Curation-relevant caveats.** All KT2440 [nuo](#) subunits carry broad EC 7.1.1.- and generic "NADH-quinone oxidoreductase subunit X" annotations; these are family-level, not KT2440-experimental. The transfer of specific cofactor/site assignments (FMN on NuoF, N2 on NuoB, quinone Asp on NuoCD) comes from *E. coli* and *T. thermophilus*, not from KT2440 experiments — a **strong** but homology-based transfer, given the deep conservation of the core.

## 4.2 The `nucCD` fusion — direct supporting evidence

The C/D fusion is not an annotation artifact. Structure/function work on the *E. coli* NuoD segment established it as the homolog of the mitochondrial 49 kDa subunit (NDUFS2) and localised the conserved quinone-site aspartate to NuoCD (PMID: 25545070; PMID: 39262040). The KT2440 "subunit C/D" annotation for PP\_4121 is therefore biochemically well-founded.

## 4.3 Genes that must NOT be mapped to Complex I core

| Gene                                                                                                                                                               | Locus   | Why excluded                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ndh</code>                                                                                                                                                   | PP_0626 | <b>Type-II NADH dehydrogenase (NDH-2)</b> — single-subunit, non-homologous to Nuo, does <b>not</b> pump protons, no Fe-S/FMN wire. A parallel/alternative NADH entry point, not a Complex I subunit. |
| PP_2867                                                                                                                                                            | Q88IY2  | Broad "pyridine nucleotide-disulphide oxidoreductase" flavoprotein family; not a Nuo subunit. Likely bucket noise.                                                                                   |
| <code>sdhA-D</code> , <code>petA-C</code> ,<br><code>cox</code> / <code>cta</code> , <code>cyo</code> ,<br><code>cco</code> , <code>cio</code> , <code>atp*</code> | various | Complexes II–V and terminal oxidases; separate modules.                                                                                                                                              |

## 4.4 Accessory subunits (out-of-taxon)

NDUFS4, NDUFS6, NDUFA9 have **no candidate gene** and **no expected bacterial ortholog**. Bacterial NDH-1 is the ~14-subunit minimal enzyme (PMID: 26807915) versus the 44-subunit mammalian complex (PMID: 23836892); the ~30 supernumerary subunits (including these three) are eukaryotic-lineage additions. Their absence is the correct null result, not a gap.

# 5. Gaps, Ambiguities, and Likely Over-Annotations

## 5.1 Not true gaps — expected absences

- **NDUFS4, NDUFS6, NDUFA9:** `not_expected_in_target_taxon`. Marking these as gaps would be a false negative driven by an eukaryote-centric module boundary.

## 5.2 Genuine curation issues

1. **Module boundary is eukaryote-centric.** The generic module expects `NDUF*` gene names and separate N-/Q-module accessory subunits that do not exist in bacteria. The module needs a **bacterial variant** keyed on `nuo` genes → `module_needs_revision`.
2. **nuoCD fusion.** Two module steps (NDUFS2, NDUFS3) satisfied by one gene (PP\_4121). The satisfiability logic must accept a single fused gene covering two steps, or the step count will look artificially short by one.
3. **KEGG bucket over-inclusion.** `ppu00190` conflates all five complexes; ~40 of 54 candidates are irrelevant to Complex I. Do not use the bucket as a Complex I gene set.
4. **NDH-2 mis-mapping risk.** `ndh` (PP\_0626) is annotated "NADH dehydrogenase (EC 1.6.99.3)" and sits in the same bucket. Automated tools may over-propagate it to a Complex I step; it must be explicitly excluded.
5. **Broad EC/GO on `nuo` subunits.** EC 7.1.1.- and generic GO ("NADH dehydrogenase activity", "electron transfer activity") are family-level; specific cofactor-binding GO (FMN binding, [4Fe-4S] binding, ubiquinone binding) is transferred by homology, not KT2440 experiment.

## 5.3 Over-annotation summary

- No over-propagation *within* the true `nuo` set — those annotations are appropriate.
- The over-annotation risk is at the **bucket level** (whole OXPHOS grouped as one map) and the **NDH-2 confounder**.

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## 6. Module and GO-Curation Recommendations

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### Per-step calls

| Step (mito subunit) | KT2440 mapping  | Call                         |
|---------------------|-----------------|------------------------------|
| NDUFV1              | nuoF / PP_4123  | covered                      |
| NDUFV2              | nuoE / PP_4122  | covered                      |
| NDUFS1              | nuoG / PP_4124  | covered                      |
| NDUFS7 (PSST)       | nuoB / PP_4120  | covered                      |
| NDUFS8 (TYKY)       | nuoI / PP_4126  | covered                      |
| NDUFS2 (49 kDa)     | nuoCD / PP_4121 | covered (fused)              |
| NDUFS3 (30 kDa)     | nuoCD / PP_4121 | covered (fused)              |
| NDUFS6              | —               | not_expected_in_target_taxon |
| NDUFA9              | —               | not_expected_in_target_taxon |
| NDUFS4              | —               | not_expected_in_target_taxon |

**Overall module status:** `module_needs_revision` — the catalytic core is genuinely covered, but the module document must be re-expressed for bacteria.

**Recommended actions** 1. Create/attach a **bacterial NDH-1 (nuo)** variant of the module that maps the seven catalytic core steps onto NuoF/E/G/B/I and fused NuoCD, and explicitly declares NDUFS4/S6/A9 as eukaryote-specific (not\_expected). 2. Add a **fusion rule** so PP\_4121 (nuoCD) satisfies both the NDUFS2 and NDUFS3 steps. 3. Add an **exclusion note** that `ndh`/PP\_0626 (NDH-2) and PP\_2867 are *not* Complex I core and must not satisfy any step. 4. Add a **bucket caveat** that KEGG `ppu00190` spans all OXPHOS complexes. 5. GO curation: retain family-level GO on `nuo` subunits; where specific cofactor GO is asserted (e.g., GO:0051539 4Fe-4S binding on nuoB/nuoI; FMN binding on nuoF), annotate with an **ISS/IEA homology evidence code**, not experimental, since no KT2440-specific biochemistry was found. No new GO term request appears necessary — existing bacterial NDH-1 GO terms are adequate.

## 7. Genes to Promote to Full fetch-gene Review

| Priority | Gene                                     | Locus               | Reason                                                                                                                      |
|----------|------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| High     | <span>nuoCD</span>                       | PP_4121             | Fused two-in-one subunit; verify fusion boundary and that it truly covers both NDUFS2 and NDUFS3; confirm quinone-site Asp. |
| High     | <span>nuoF</span>                        | PP_4123             | N-module catalytic entry (FMN/NADH, ~NDUFV1); anchor of the N-module call.                                                  |
| High     | <span>nuoB</span>                        | PP_4120             | Terminal N2 [4Fe-4S] cluster (~NDUFS7/PSST); electron donor to quinone.                                                     |
| Medium   | <span>nuoI</span>                        | PP_4126             | TYKY/NDUFS8 [4Fe-4S] wire; confirm both cluster-binding motifs.                                                             |
| Medium   | <span>nuoE</span> ,<br><span>nuoG</span> | PP_4122,<br>PP_4124 | N1a and 75 kDa Fe-S relay; confirm cluster complement.                                                                      |
| Medium   | <span>ndh</span>                         | PP_0626             | Confirm NDH-2 identity and <b>explicitly exclude</b> from Complex I to prevent auto over-propagation.                       |

## 8. Mechanistic Model / Interpretation

*P. putida* KT2440 runs an aerobic, branched respiratory chain in which NADH oxidation is served by **two independent, non-homologous enzymes**: the proton-pumping **Complex I / NDH-1** (nuo, PP\_4119–4131) and the non-pumping **NDH-2** (ndh, PP\_0626). Only the former corresponds to the reviewed mitochondrial Complex I core. Its peripheral arm is a linear electron conduit that is structurally and mechanistically conserved across all domains of life:

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NADH → FMN(NuoF/NDUFV1) → N3 → N1b → N4 → N5(NuoG/NDUFS1)
      → N6a/N6b(NuoI/NDUFS8) → N2(NuoB/NDUFS7) → Quinone
                                |
                                quinone cavity shaped by fused NuoCD (NDUFS2+NDUFS3)

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The quinone chemistry at the deep end of the cavity — protonation of a conserved aspartate near cluster N2 — has been pinned to the NuoCD/49-kDa subunit and is central to redox-coupled proton pumping ([PMID: 39262040](#); [PMID: 26330610](#); [PMID: 23417064](#)). Because this machinery is conserved and KT2440 carries a **single, complete, syntenic** operon, the module is satisfiable with high confidence, even without strain-specific mutant data. The eukaryotic accessory subunits (NDUFS4/S6/A9) are structural/assembly add-ons of the larger mitochondrial complex; the bacterial enzyme achieves the same catalysis without them, which is why they are legitimately absent here.

## 9. Evidence Base

| PMID                                                                                                            | How it supports the review                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">25545070</a>                                                                                        | <i>"The NuoD segment (homologue of mitochondrial 49 kDa subunit)"</i> — establishes NuoD = NDUFS2, underpinning the Q-module core mapping and the nuoCD fusion.                                                  |
| <a href="#">39262040</a>                                                                                        | <i>"a specific, conserved aspartic acid residue in the quinone binding site (D325 on subunit NuoCD in Escherichia coli)"</i> — confirms the fused NuoCD polypeptide and the quinone-cavity core (NDUFS2/NDUFS3). |
| <a href="#">26807915</a>                                                                                        | <i>"Bacterial enzyme is about half the size of mitochondrial and thus provides its important 'minimal' model."</i> — supports that NDUFS4/S6/A9 accessory subunits are not expected in bacteria.                 |
| <a href="#">23836892</a>                                                                                        | <i>"Complex I ... in mammalian mitochondria is an L-shaped assembly of 44 protein subunits"</i> — contrasts 44-subunit mammalian vs ~14-subunit bacterial core.                                                  |
| <a href="#">23417064</a>                                                                                        | Crystal structure of intact <i>T. thermophilus</i> Complex I — defines the conserved 16-subunit core, quinone chamber and N2 cluster used for the mapping.                                                       |
| <a href="#">41977177</a>                                                                                        | <i>"This basic core subunit architecture is highly conserved from bacterial to mammalian CI"</i> — justifies homology transfer to KT2440.                                                                        |
| <a href="#">26330610</a> ,<br><a href="#">30697773</a> ,<br><a href="#">32347721</a> , <a href="#">24973951</a> | Mechanistic/structural detail on Fe-S wire, cluster potentials, quinone site and membrane-arm proton pumping — supports the conserved electron-transfer logic.                                                   |

**Evidence class:** Homology/orthology (UniProt/KEGG assignments) with **strong biochemical transfer** from *E. coli* and *T. thermophilus* Complex I. No *P. putida* KT2440-specific Complex I subunit-mutant or structural study was found in the literature searched; the target-organism evidence is genomic (single complete syntenic `nuo` operon) plus annotation, not direct enzymology.

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## 10. Limitations and Knowledge Gaps

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1. **No KT2440-specific functional data.** All catalytic/cofactor assignments are transferred by homology from *E. coli/Thermus*. Transfer is strong (deep conservation) but not experimentally verified in KT2440.
  2. **Native quinone identity.** The physiological quinone (ubiquinone-9 vs menaquinone) and its exact affinity in KT2440 were not established here; this affects downstream OXPHOS reasoning but not the core mapping.
  3. **NDH-2 vs NDH-1 flux partitioning.** KT2440 uses both NADH dehydrogenases; the relative in vivo contribution is condition-dependent and out of scope but relevant to any energetic module.
  4. **Cofactor stoichiometry per subunit** (exact Fe-S cluster complement of `nuoG/nuoI` in KT2440) is inferred, not measured.
  5. **Assembly factors.** Bacterial NDH-1 assembly factors were not surveyed; the module scope was the structural core only.
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## 11. Proposed Follow-up Actions

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1. **Promote PP\_4121 (`nuoCD`), PP\_4123 (`nuoF`) and PP\_4120 (`nuoB`) to full `fetch-gene` review** to confirm the fusion boundary, FMN/NADH site and N2 cluster respectively.
2. **Author a bacterial NDH-1 module variant** mapping the seven core steps onto `nuo` genes, encoding the `nuoCD`-covers-two-steps rule, and declaring `NDUFS4/S6/A9` as `not_expected_in_target_taxon`.
3. **Add exclusion flags** for `ndh`/PP\_0626 (NDH-2) and PP\_2867 so they cannot satisfy Complex I steps.
4. **Add a KEGG bucket caveat** documenting that `ppu00190` spans all five OXPHOS complexes.

**5. Optional experimental resolution:** targeted deletion or complementation of the KT2440 `nuo` operon, or purified-enzyme assays, would convert the current homology-based calls to direct evidence — but is likely unnecessary for satisfiability given the completeness and synteny of the locus.

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## 12. Key References

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- Efremov & Sazanov. *Structure of bacterial respiratory complex I*. PMID: 26807915
  - Baradaran et al. *Crystal structure of the entire respiratory complex I*. PMID: 23417064
  - *Conserved amino acid residues of the NuoD segment important for structure and function of E. coli NDH-1 (complex I)*. PMID: 25545070
  - *Quinone chemistry in respiratory complex I involves protonation of a conserved aspartic acid residue*. PMID: 39262040
  - *Post-translational modifications near the quinone binding site of mammalian complex I*. PMID: 23836892
  - *Roles of Subunit ND2/NuoN in the Proton Pumping Coupling Mechanism of Complex I*. PMID: 41977177
  - *Redox-induced activation of the proton pump in the respiratory complex I*. PMID: 26330610
  - *Reduction potential calculations of the Fe-S clusters in T. thermophilus respiratory complex I*. PMID: 30697773
  - *Charge Transfer and Chemo-Mechanical Coupling in Respiratory Complex I*. PMID: 32347721
  - *Essential regions in the membrane domain of bacterial complex I (NDH-1)*. PMID: 24973951
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*Prepared for manual module satisfiability and gene-annotation curation. Target organism: Pseudomonas putida KT2440 (taxon 160488). Evidence base: 21 papers reviewed, 5 findings confirmed across 3 iterations.*

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